



Internship Project — Week 1

Project Title: Sentiment Analysis on Product Reviews

Assigned Date: 28/04/2026 **Submission Date:** 3/05/2026



Problem Statement

E-commerce platforms receive thousands of customer reviews every day. Manually reading each review to understand customer satisfaction is time-consuming and inefficient. Your task is to build a **Sentiment Analysis system** that automatically reads product reviews and classifies them as **Positive, Negative, or Neutral** — and then generates meaningful insights from the results.

You will work with a real-world product reviews dataset, clean and process the text data, apply a sentiment analysis model, and present your findings through charts and a short report.



Dataset — Where to Get It

Go to this link and download the dataset:



<https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>

Steps to download:

1. Create a free account on www.kaggle.com (if you don't have one)
2. Search for **"Amazon Fine Food Reviews"**
3. Download the file — it will be a **.csv** file named **Reviews.csv**
4. Use the **first 5,000 rows only** to keep it manageable



Tasks to Complete

Complete all the following tasks in a **Jupyter Notebook (.ipynb)**:

Task 1 — Data Loading & Exploration

- Load the CSV file using Pandas
- Display the first 10 rows
- Check how many rows and columns are there
- Identify which column contains the review text

Task 2 — Data Cleaning

- Remove rows where the review text is empty or null
- Remove duplicate reviews
- Keep only the columns you need: review text + rating (score)

Task 3 — Sentiment Analysis

- Install and use the **TextBlob** library (`pip install textblob`)
- For each review, calculate the **sentiment polarity score**
- Based on the score, label each review as:
 - **Positive** → score > 0
 - **Negative** → score < 0
 - **Neutral** → score = 0

Task 4 — Visualization (Minimum 3 charts)

- **Chart 1:** Bar chart showing count of Positive, Negative, and Neutral reviews
- **Chart 2:** Pie chart showing the percentage distribution of sentiments
- **Chart 3:** A chart of your own choice — be creative! (e.g., *most common words in negative reviews*, *rating vs sentiment comparison*, etc.)

Task 5 — Insights & Summary Write a short paragraph (5–8 lines) inside your notebook answering:

- What percentage of reviews are positive?
- What do customers mostly complain about in negative reviews?
- What surprised you in the data?
- One recommendation for the business based on your findings

Tools & Libraries to Use

Tool	Purpose
Python 3.x	Main programming language
Jupyter Notebook	To write and run your code
Pandas	Data loading and cleaning
TextBlob	Sentiment analysis
Matplotlib / Seaborn	Charts and visualization

What to Submit

Create a folder named: `SentimentAnalysis_[YourName]`

Inside the folder, include:

-  `analysis.ipynb` — Your complete Jupyter Notebook with all 5 tasks
-  `Reviews.csv` — The dataset you used (*first 5,000 rows only*)
-  `summary.pdf` or `summary.docx` — A 1-page written summary of your findings (can be exported from the notebook)
-  `charts/` — A folder with all 3 chart images saved as `.png`

Submit the entire folder as a ZIP file

Via:-

<https://docs.google.com/forms/d/e/1FAIpQLSf7t2YhRzjpQi0CL4TtG6gjqGWZ168Rk047SmuPTAXIUB7Flw/viewform?usp=dialog>